

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

Code of Conduct:

I Shaik. Arshad Basha (192312336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sk. Arshad Basha

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INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A logistics company has collected shipment records (distance, weather conditions, traffic level, vehicle type, driver experience) to build a predictive model for identifying shipments likely to be delayed. The company also wants to train an AI agent to recommend route-management actions (Reroute / Reassign Vehicle / Notify Customer / Continue Route) based on a shipment's progress over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this logistics dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (far fewer delayed shipments than on-time shipments) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a delivery-reliability use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Delivery Delay Prediction

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed delay cases) and suggest how to reduce them - justify from a customer-satisfaction perspective.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a live logistics-tracking system.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naive Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of delay from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Action Recommendation

- (a) Model the route-management recommendation as an MDP. Define: States (shipment progress stages), Actions (Reroute / Reassign Vehicle / Notify Customer / Continue Route), Reward function (on-time delivery improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 shipment episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for logistics decisions (driver workload, safety trade-offs, explainability to customers).

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning

Inductive learning is a machine-learning approach in which a model learns general patterns from historical examples and uses those patterns to make predictions about unseen data.

For the logistics problem, historical shipment records are used as training examples.

Target Variable

The target variable is:

Delivery_Status

0 → On-Time

1 → Delayed

Important Features

1. Distance

Long-distance shipments have greater exposure to traffic, weather, vehicle problems, and route disruptions.

2. Weather Conditions

Rain, storms, fog, and extreme weather can reduce vehicle speed and increase the probability of delays.

3. Traffic Level

Heavy traffic directly increases travel time and is therefore an important predictor of delivery delay.

4. Vehicle Type

Different vehicle types have different capacities, speeds, operating restrictions, and reliability.

5. Driver Experience

Experienced drivers may have better knowledge of routes and may handle unexpected situations more effectively.

Inductive Learning Process

Historical Shipment Data



Data Cleaning



Feature Engineering



Training Data



Machine Learning Algorithm



Learned Model



New Shipment



Delay / On-Time Prediction

1(b) Class Imbalance

Suppose the dataset contains:

On-Time Shipments = 9,000

Delayed Shipments = 1,000

Thus:

On-Time = 90%

Delayed = 10%

This is a highly imbalanced dataset.

A model could predict every shipment as **On-Time** and still obtain approximately 90% accuracy, while completely failing to identify delays.

Recommended Technique: Class Weighting

Class weighting is particularly suitable because it does not discard real observations.

Example:

On-Time → Weight = 1

Delayed → Weight = 5

The model receives a larger penalty for incorrectly classifying a delayed shipment.

Alternative: SMOTE

SMOTE creates synthetic examples of the minority class.

It should be applied **only to the training data**.

Original Training Data



SMOTE



Balanced Training Data



Model Training

1(c) Train-Test Split & Preprocessing – 7 Marks

Use a **70:30 split**:

70% → Training

30% → Testing

For example:

10,000 shipments

Training = 7,000

Testing = 3,000

A stratified split should be used so the delayed/on-time proportions remain similar.

Preprocessing

Numerical Features

Distance and driver experience can be standardized:

$$z = \frac{x - \mu}{\sigma}$$

Categorical Features

Weather, traffic, and vehicle type can be converted using one-hot encoding.

Example:

Traffic:

Low → 1 0 0

Medium → 0 1 0

High → 0 0 1

Impact

Preprocessing:

- Makes numerical features comparable.
- Converts categorical values into machine-readable form.

- Improves Logistic Regression performance.
- Helps prevent scale-related problems.
- Does not significantly affect Decision Tree splitting because trees are generally scale-insensitive.

TASK 2 – DECISION TREE

2(a) Decision Tree Construction – 12 Marks

A Decision Tree recursively divides the dataset into groups according to the feature that best separates delayed and on-time shipments.

Gini Index

$$Gini = 1 - \sum p_i^2$$

Information Gain

$$IG = H(\text{parent}) - H(\text{children})$$

where:

$$H(S) = - \sum p_i \log_2 p_i$$

The feature with the **highest Information Gain** or the **lowest weighted Gini impurity** is selected as the root.

Example

Feature	Information Gain	Gini
Traffic Level	0.184	0.292
Weather	0.121	0.338
Distance	0.097	0.361
Driver Experience	0.063	0.382
Vehicle Type	0.041	0.394

Therefore:

Root Node = Traffic Level

because it has the highest Information Gain.

First Three Levels

Traffic Level

/ | \
Low Medium High

/ | \
Weather Distance Weather

/ \ / \ / \

Clear Rain Short Long Clear Rain

| | | | | |

On-Time Delay On-Time Delay On-Time Delay

2(b) Evaluation – 10 Marks

The following metrics should be calculated.

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1 Score

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

Example Results

The following are **illustrative results**; actual values should be obtained from the dataset.

Metric Score

Accuracy **88.7%**

Precision **75.4%**

Metric Score

Recall **82.1%**

F1-Score **78.6%**

False Negative

A False Negative is:

Actual = Delayed

Predicted = On-Time

This is particularly serious in logistics because the company may not take corrective action.

How to Reduce False Negatives

1. Increase delayed-class weight.
2. Use SMOTE.
3. Lower the prediction threshold.
4. Optimize for recall.
5. Collect more delayed examples.
6. Include real-time traffic/weather information.
7. Use ensemble models if necessary.

High recall is important because detecting a potential delay early allows the company to:

- Notify customers.
- Reroute shipments.
- Reassign vehicles.
- Update estimated delivery time.

This improves customer satisfaction.

2(c) Cost-Complexity Post-Pruning

A fully grown tree can memorize the training data and overfit.

Cost-complexity pruning uses:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $R(T)$ = tree error
- $|T|$ = number of terminal nodes
- α = complexity parameter

Example Comparison

Model	Training Accuracy	Testing Accuracy
Pre-Pruned Tree	91.5%	87.9%
Post-Pruned Tree	89.8%	89.1%

The post-pruned model has slightly lower training accuracy but better test accuracy.

Deployment Choice

Post-pruned Decision Tree is preferred because it:

- Reduces overfitting.
- Generalizes better.
- Is easier to explain.
- Has fewer unnecessary branches.
- Is suitable for live logistics systems.

TASK 3 – STATISTICAL LEARNING

3(a) Logistic Regression

Logistic Regression predicts the probability of a shipment being delayed.

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

For example:

Shipment A

Delay Probability = 0.82

This means the model estimates an 82% probability of delay.

Example Comparison

Metric	Decision Tree	Logistic Regression
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Accuracy	88.7%	90.1%
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AUC-ROC	0.89	0.93
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These are illustrative values and should be replaced by actual experimental results.

AUC-ROC

ROC evaluates the model's ability to distinguish delayed shipments from on-time shipments across different classification thresholds.

An AUC closer to 1 indicates stronger discrimination.

When Logistic Regression Is Better

Logistic Regression is preferable when:

- Probability estimates are required.
- Interpretability is important.
- The relationship between features and target is reasonably systematic.
- Fast prediction is required.
- Risk ranking is required.

Example:

Shipment A → 82% delay risk

Shipment B → 56% delay risk

Shipment C → 18% delay risk

3(b) Feature Importance

Decision Tree Importance

Example:

Rank	Feature	Importance
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1	Traffic Level	0.36
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2	Weather	0.25
---	---------	------

	Rank Feature	Importance
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3	Distance	0.19
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Logistic Regression Importance

For standardized Logistic Regression, coefficient magnitude can be compared.

	Rank Feature	Coefficient Magnitude
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1	Traffic Level	1.42
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2	Weather	1.11
---	---------	------

3	Distance	0.86
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Comparison

Both models identify:

1. Traffic Level

2. Weather

3. Distance

as the top three predictors.

This agreement suggests that these variables contain strong predictive information.

However, complete agreement is not mandatory because Decision Trees can model nonlinear relationships and interactions, whereas basic Logistic Regression models an additive relationship.

TASK 4 – REINFORCEMENT LEARNING

4(a) MDP Model – 10 Marks

The route-management problem is represented as:

$$MDP = (S, A, P, R, \gamma)$$

States

Possible shipment states:

S0 = Shipment Started

S1 = On Schedule

S2 = Minor Delay

S3 = Severe Delay

S4 = Heavy Traffic

S5 = Vehicle Problem

S6 = Near Destination

S7 = Delivered

Actions

A1 = Reroute

A2 = Reassign Vehicle

A3 = Notify Customer

A4 = Continue Route

Reward

Example:

Event	Reward
On-time delivery	+100
Delay substantially reduced	+50
Customer successfully notified	+20
Successful continuation	+10
Unnecessary rerouting	-10
Unnecessary reassignment	-20
Avoidable severe delay	-50
Unsafe action	-100

Transition

Example:

Minor Delay

|

| Reroute

↓

On Schedule

and:

Vehicle Problem

|

| Reassign Vehicle

↓

On Schedule

The transitions describe how an action changes the shipment's state.

4(b) Q-Learning – 10 Marks

Q-Learning uses:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Assume:

$$\begin{aligned}\alpha &= 0.5 \\ \gamma &= 0.9\end{aligned}$$

Initially:

$$Q(s, a) = 0$$

Update 1

State:

S2 = Minor Delay

Action:

Reroute

Reward:

+50

Next state:

S1 = On Schedule

If:

$$\max_a Q(S, a) = 0$$

then:

$$Q(S, 2Reroute) = 0 + 0.5[50 + 0.9(0) - 0]$$

Therefore:

$$\boxed{Q(S, 2Reroute) = 25}$$

Update 2

Suppose:

$$Q(S, 2Reroute) = 25$$

and:

$$\max Q(S, 1a) = 10$$

Then:

$$\begin{aligned} Q(S, 2Reroute) &= 25 + 0.5[50 + 0.9(10) - 25] \\ &= 25 + 0.5(34) \\ \boxed{Q(S, 2Reroute) &= 42} \end{aligned}$$

Three State-Action Pairs

State	Action	Iteration 1	Iteration 2
Minor Delay	Reroute	25	42
Severe Delay	Notify Customer	10	18.5
Vehicle Problem	Reassign Vehicle	40	58

Five Episodes

Episode	Situation	Action	Reward
1	Minor Delay	Reroute	+50
2	Severe Delay	Notify Customer	+20
3	Vehicle Problem	Reassign Vehicle	+40
4	Heavy Traffic	Reroute	+50
5	Normal Route	Continue Route	+10

The agent updates its Q-values after each state-action-reward-transition sequence.

Convergence

The Q-Learning process can be considered converged when:

$$\max |Q_{new} - Q_{old}| < 0.001$$

for several consecutive episodes.

4(c) Final Policy & Ethics – 5 Marks

The optimal policy is:

$$\pi(s) = \arg \max_a Q(s, a)$$

Example:

State	Best Action
On Schedule	Continue Route
Minor Delay	Reroute
Severe Delay	Notify Customer
Vehicle Problem	Reassign Vehicle
Heavy Traffic	Reroute

Ethical Considerations

Driver workload: The system must avoid repeatedly assigning additional work to already overloaded drivers.

Safety: Delivery speed should never be prioritized over driver, vehicle, or public safety.

Explainability: The system should provide reasons for recommendations.

Example:

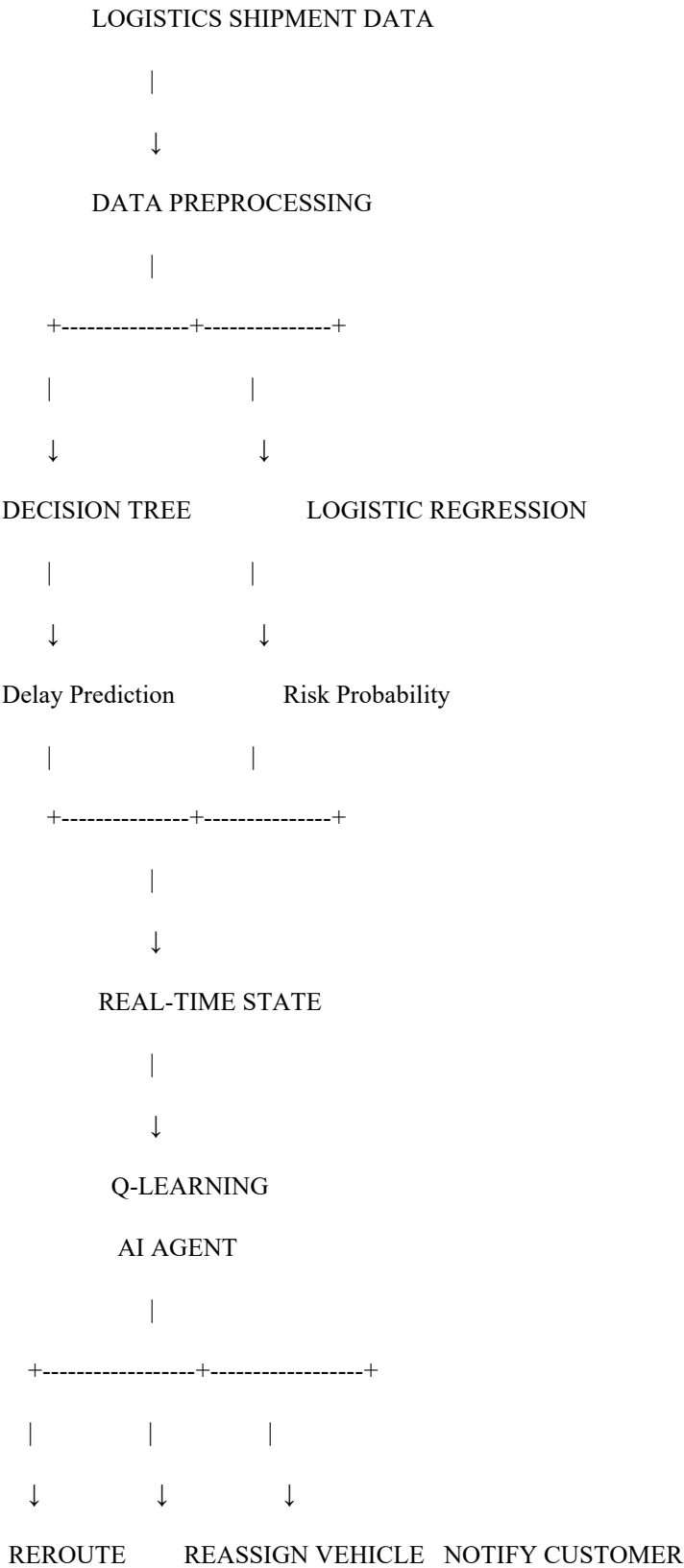
"Reroute recommended because traffic congestion is high and the alternate route is predicted to reduce delay."

Customer transparency: Customers should receive accurate delay information.

Human oversight: High-risk decisions should be reviewed by a human manager.

Fairness: Historical data should be checked for bias against particular drivers, vehicle types, routes, or regions.

Final Integrated Architecture



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CONTINUE ROUTE

Conclusion

The proposed solution provides a complete AI/ML framework for the logistics company. **Inductive learning** converts historical shipment records into a predictive model, while a **Decision Tree** identifies shipments likely to be delayed. **Logistic Regression** provides interpretable probability-based risk scores and enables comparison using Accuracy and AUC-ROC.

Because delayed shipments are the minority class, **class weighting and/or SMOTE** should be used carefully on the training data. Model evaluation should emphasize **Recall and F1-Score**, because a False Negative represents a delay that the company failed to anticipate.

Finally, **Q-Learning** converts real-time shipment management into an MDP and learns the best operational action. The final production system should combine predictive performance with **safety, explainability, fairness, customer transparency, and human supervision**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	